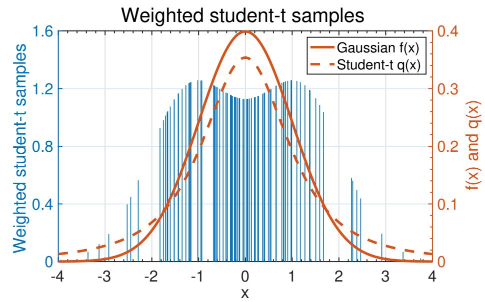
## Humans don't understand sums of impulses

- We are nearly ready to put all the pieces together to develop the particle filter. But, one issue remains.
- You have learned that we represent pdfs as sums of Dirac delta impulse functions.
- We would like to be able to visualize particle estimates of a pdf, especially $\hat{f}(x_k \mid \mathbb{Z}_k)$.
- But, it's difficult to decode what the pdf must look like simply by looking at a plot of weighted impulses, as the figure from a prior example shows.
- That is, the weighted impulses store information in a way that is difficult to grasp intuitively.








## Kernel functions

- A popular method of visualizing a particle-based pdf is to add a series of "bumps" together, where each bump is scaled by the impulse weight and centered at the impulse location.
  - This is merely popular; it is not necessarily "best."
- The bumps are called kernels and should have the following properties

$$b(u) \geq 0, \quad \text{and} \quad \int_{-\infty}^{\infty} b(u)\,\mathrm{d}u = 1.$$

- The width of the kernel is specified by $\sigma$. Typically

$$b_\sigma(u) = \frac{1}{\sigma} b\left(\frac{u}{\sigma}\right),$$

where it is easy to show that $\int b_\sigma(u)\,\mathrm{d}u = \int b(u)\,\mathrm{d}u = 1$.



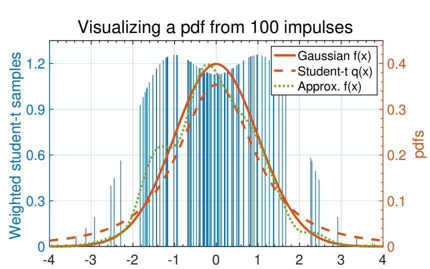


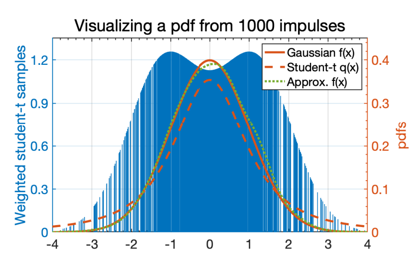
## Visualizing a pdf using kernels

- Then, a kernel density estimator is simply expressed as:

$$\hat{f}(x_k) = \frac{1}{N_p} \sum_{i=1}^{N_p} w_k^{(i)} b_\sigma(x_k - x_k^{(i)}).$$

- The impulse function $\delta(\cdot)$ is replaced by the scaled "bump" function $b_\sigma(\cdot)$ in the sum.
- For the prior example, we can visualize the pdf using Gaussian-shaped kernels as:



## Kernel options

- Kernels usually have even symmetry.
- Popular kernels include:

  | | |
  |---|---|
  | Epanechnikov: | $b(u) = c(1-u^2)\Pi(u/2)$ |
  | Bi-weight: | $b(u) = c(1-u^2)^2\Pi(u/2)$ |
  | Tri-weight: | $b(u) = c(1-u^2)^3\Pi(u/2)$ |
  | Triangular: | $b(u) = c(1-\lvert u\rvert)\Pi(u/2)$ |
  | Gaussian: | $b(u) = ce^{-u^2}$ |

  where $c$ is a constant chosen to make the kernel integrate to 1, and $\Pi(u)$ is the unit-pulse function: $\Pi(u) = 1$ for $|u| < 0.5$; else, $\Pi(u) = 0$.
- Bump width is the critical parameter. Much more important than the kernel choice.
  - $\sigma$ large: Smoother pdf estimate, more biased, less variable;
  - $\sigma$ small: Coarser pdf estimate, less biased, more variable.
- Quiz introduces "Hellinger distance" to enable judging quality of approximation.





## Summary

- Representing pdfs as a sum of impulses is efficient from a computational point of view but they are difficult to visualize.
- To assist intuition, we can replace the impulses in the summation that represents a pdf with a kernel "bump" function.
- If the width of the bump function is well chosen, plotting this summation of bump functions will approximate the true pdf being modeled by the sum of impulses.
- Bumps should have unit area; they are often symmetric.
- Their width is generally more important than their specific shape:
  - Bumps that are "too wide" lose detail and smear the approximate pdf;
  - Bumps that are "too narrow" look very much like the impulses themselves and do not assist with interpreting the pdf.
- Some trial-and-error may be needed to use this method successfully, but it can be very helpful in the end to aid intuition.